

VOLUME 1

COMPLETE VOLUME — CHAPTERS 1.1–1.11

Product Planning

Defining Exactly What the Product Is

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This volume defines exactly what the Human Archive Collector app must be, grounded in the product flow and feature list already confirmed by the project owner. It contains the following 11 chapters:

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VOLUME 1 — PRODUCT PLANNING

CHAPTER 1.1

Product Requirements Document

PRD — What the Product Must Do and Why

Document Title	Product Requirements Document (PRD)
Chapter Reference	1.1
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Prepared With	Claude (Cowork)
Date	05 August 2026
Revision Note	v1.1 — Promoted Admin from a future phase to an MVP-scope role, and expanded metadata capture into a first-class core requirement, per owner request.
Governed By	Volume 0 — Project Foundation (Vision Document, Project Constitution, Glossary)

1. Overview

This PRD defines exactly what the Human Archive data-collection platform must do. It is grounded directly in the workflow and feature list already defined by the project owner, and expands each element into a reviewable requirement. It does not introduce new product ideas beyond what has already been specified — where a gap exists, it is marked as an open question rather than invented.

As of this revision, the platform is understood to have two roles from day one: an Admin who creates and manages Projects, Tasks, and Collector assignments, and a Collector (User) who views assigned work and records/uploads footage against it. Every chunk and session the platform captures must also carry complete, verifiable metadata — this is now treated as a core requirement, not an incidental detail of chunking.

2. Problem & Solution Summary

Problem: Field data collection for egocentric, wide-angle video today relies on ad-hoc recording apps, manual file handling, and no structured link between footage and the task it belongs to — causing inconsistent capture, data loss, no traceability, and no resilience to network interruption (see Volume 0, Chapter 0.1, Section 1). There is also no one to structure the work itself — someone must be able to define Projects and Tasks and hand them to the right Collector.

Solution: A dedicated Flutter data-collection platform with two roles. An Admin creates Projects and Tasks and assigns Collectors to them. A Collector logs in, sees only what's assigned to them, passes a pre-recording health check, records wide-angle rear-facing footage, and lets the app automatically chunk and upload it in the background — while every chunk is accompanied by a complete, tamper-evident metadata record from the moment it is created.

3. Roles at a Glance

Full role and permission detail lives in Chapter 1.6 — User Roles & Permissions. In summary, for the MVP:

Admin: Creates and edits Projects and Tasks (the "lists" of work), assigns Collectors to them, and can view the status, footage, and metadata of everything under their Projects.

Collector (User): Sees only the Projects/Tasks assigned to them, records against a Task, and lets the app handle chunking, metadata generation, and upload.

A Super Admin role (managing Admins themselves and client-level configuration) remains a future-phase concept and is not required for MVP — see Chapter 1.6.

4. End-to-End Product Flow

4.1 Collector Flow

This is the confirmed screen-to-screen flow the Collector side of the product must support, exactly as specified:

1. Install App
2. Onboarding / Permissions Priming
3. Login (Email/Password or SSO)
4. Home Dashboard
5. Assigned Projects List
6. Project Detail → Task List
7. Task Detail (instructions, examples, requirements)
8. Pre-Recording Checklist (permissions, storage, battery, network)
9. Recording Screen (live capture)
10. Stop Recording
11. Automatic Chunking (splits into 10-minute files locally)
12. Local Chunk Processing (including metadata generation — Section 6)
13. Automatic Upload begins (or manual upload if disabled)
14. Upload Progress / Sync Status
15. Session Marked Complete
16. Return to Task List / Dashboard

4.2 Admin Flow (New — MVP Scope)

This flow did not exist in the original Collector-only spec and is added here because Admin capability is now required for the MVP. Exact screens/surface (mobile, web, or both) are an open question — see Section 8.

17. Admin logs in.
18. Admin Dashboard — shows all Projects the Admin manages, and a summary of Collector activity across them.
19. Create/Edit Project.
20. Create/Edit Task within a Project (instructions, examples, requirements).
21. Assign one or more Collectors to a Project and/or specific Tasks.
22. View Session/Chunk status and metadata for any Task under their Projects.
23. (Optional, may phase in) Reassign or remove a Collector from a Task.

5. Key Features

5.1 Core Features (MVP — see Chapter 1.10)

- Secure login (credentials + SSO), session/token management, for both Admin and Collector.
- Admin: create/edit Projects and Tasks, and assign Collectors to them.
- Collector: Project + Task assignment retrieval, scoped strictly to what they are assigned.
- Rear-facing continuous, high-bitrate, wide-angle (0.5x–0.6x) video recording.
- Automatic 10-minute segment chunking on stop (not during recording).
- Local chunk persistence immediately after stop.
- Automatic or manual upload mode for chunks, configurable per device.
- Background, resumable, chunked upload directly to S3.
- Full per-chunk and per-session metadata capture, integrity verification, and storage (see Section 6).
- Offline recording and upload retry (chunks queue locally).
- Session lifecycle management.
- Pre-recording device health checklist (storage, battery, permissions, network).
- Upload status visibility (dashboard + per-session), visible to both Collector (own work) and Admin (their Projects).

5.2 Secondary Features

- Session notes/tagging.
- Task instructions with reference media.
- Push notifications (upload failures, new task assignments).
- Multi-project support per Collector.
- Local caching of task/project data for offline browsing.
- In-app support/help contact.
- Manual upload retry and pause/resume controls.

5.3 Advanced Admin & Ops Features (Phase 3)

- Live upload/collection monitoring dashboard across the fleet.
- Dataset QA/review and rework flagging.
- Collector performance and throughput analytics.
- Remote configuration (chunk duration, resolution, bitrate) via feature flags.

5.4 Future Capabilities (Phase 4)

- OTA remote configuration (no app-store release needed for capture parameters).
- AI-assisted in-app QA (e.g. blur/occlusion/motion-blur detection before upload).

- Multi-language localization.
- Multi-camera support (front + rear simultaneous capture).
- Wearable capture integration.

6. Metadata — Now a Core Requirement

Every chunk and session the platform produces must be accompanied by a complete, verifiable metadata record — this applies from the MVP onward, not as a later enhancement. Full atomic requirements are in Chapter 1.3 (FR-META); the intent is captured here for product-level visibility:

- Every chunk carries an immutable, system-generated metadata record from the instant it is created — including identity (chunk/session/project/task/collector IDs), timing (start/end, duration, sequence index), technical capture details (resolution, bitrate, codec, frame rate, zoom factor), device/app context (device model, OS version, app version), capture conditions (GPS location if permitted, battery level, network state), and file details (size, integrity checksum).
- Metadata is persisted locally immediately, alongside the chunk itself, so it survives a crash or force-close exactly like the video does.
- Metadata travels with its chunk to the backend and is verified there (checksum match) before the chunk is considered Complete.
- Collectors can never edit system-generated metadata — only optional notes/tags they add themselves.
- Admins can view (and, where needed, export) the metadata for any chunk or session under their Projects.

7. Detailed Flow Notes (per screen)

7.1 Install & First Launch

- App requests Camera, Microphone, Location (When In Use), Notifications, and Files access.
- If a required permission is denied, the app shows a blocking explainer screen with a settings deep link.
- The app does not allow recording without Camera + Microphone permission granted.

7.2 Login

- Email + password, or enterprise SSO (SAML/OAuth) depending on client configuration.
- Token refresh happens silently; forced re-authentication only on token expiry.
- The same login mechanism serves both Admin and Collector; the role determines what is shown afterward.

7.3 Home Dashboard

- Collector view: active projects, in-progress sessions, total recorded time, and sync status.
- Admin view: all managed Projects, Collector activity summary, and outstanding Task counts.
- Displays pending / uploading / completed chunk counts.

7.4–7.6 Projects → Tasks → Task Detail

- Standard hierarchical drill-down: Project → Task List → Task Detail, consistent regardless of platform, scoped by role (Section 3).

7.7 Recording Screen

- Full-screen camera preview using the rear-facing wide lens (Android) / ultra-wide lens (iPhone 11+).
- Record/Stop button, elapsed timer, recording indicator.
- Tapping Stop triggers local chunking and metadata generation — neither happens mid-recording.

7.8 Upload Phase

- Automatic (default): chunks upload in the background as soon as they are chunked, metadata included.
- Manual: the Collector taps "Upload" to start, if automatic mode is disabled for that device.
- Upload resumes automatically when network connectivity returns.

8. Assumptions & Open Questions

The following are treated as assumptions until the project owner confirms or corrects them in review — they do not block this PRD's approval but must be resolved before the corresponding later volume (in parentheses) is finalized:

- Backend API and hosting provider are not yet finalized (Volume 4 — Backend Architecture).
- Exact SSO providers to support (e.g. Okta, Azure AD, Google Workspace) are client-dependent and not yet enumerated (Volume 4.7 — Authentication Design).
- Whether video review/preview before upload is required for the MVP is not yet confirmed (Volume 1.10 — MVP Scope).
- Target Android/iOS minimum OS versions are not yet set (Volume 6 — Mobile App Architecture).
- Whether the Admin surface is a mobile screen inside the same app, a separate web dashboard, or both for MVP is not yet decided (Volume 2 — Product Design, Volume 6 — Mobile App Architecture).

- The exact metadata storage mechanism on the backend (dedicated table vs. document store vs. S3 object tags) is not yet finalized (Volume 4.4/4.5 — Data Dictionary & Metadata Specification).

9. Out of Scope for This Document

- Detailed screen-by-screen UI specification — covered in Volume 2 (Product Design).
- Technology selection and system architecture — covered in Volume 3 (Technical Architecture) and Volume 4 (Backend Architecture).
- Detailed camera/chunking/upload engineering design — covered in Volume 5 (Recording Engine).
- Exact metadata schema/data types — covered in Volume 4.4 (Data Dictionary) and Volume 4.5 (Metadata Specification).

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CHAPTER 1.2

Software Requirements Specification

SRS — Formal Specification of System Behavior

Document Title	Software Requirements Specification (SRS)
Chapter Reference	1.2
Version	1.1
Status	Draft — Pending Approval
Owner	Faisal — Human Archive
Prepared With	Claude (Cowork)
Date	05 August 2026
Revision Note	v1.1 — Admin is now in scope as a current system feature (3.8), and metadata generation/integrity is specified as its own system feature (3.9).
Based On	Chapter 1.1 — Product Requirements Document

1. Introduction

1.1 Purpose

This SRS translates the Product Requirements Document (Chapter 1.1) into a formal specification suitable for engineering handoff. It defines what the system must do (functional behavior), how well it must do it (non-functional behavior, detailed further in Chapter 1.4), and the boundaries of the system.

1.2 Scope

The system specified here is the Human Archive data-collection platform (Flutter, Android + iOS) for both of its MVP roles — Admin and Collector — plus the minimum backend services it depends on: authentication, Project/Task management and assignment, chunked video upload to cloud storage (AWS S3), and metadata generation/storage. Advanced Admin/Ops capabilities (fleet-wide analytics, QA/rework workflow, remote configuration) remain Phase 3 and are out of scope for this SRS.

1.3 Definitions, Acronyms, and Abbreviations

All terms used in this document are defined once, authoritatively, in Volume 0, Chapter 0.3 — Glossary. This SRS does not redefine terms already covered there.

1.4 References

- Volume 0 — Project Foundation (Vision Document, Project Constitution, Glossary)
- Chapter 1.1 — Product Requirements Document
- Chapter 1.3 — Functional Requirements
- Chapter 1.4 — Non-Functional Requirements

2. Overall Description

2.1 Product Perspective

The platform is new and standalone — it does not replace or integrate with an existing system. It is a two-role, two-sided system from the MVP onward: an Admin who structures the work (Projects, Tasks, Collector assignment) and a Collector who executes it (recording, chunking, upload). Advanced Admin/Ops tooling beyond core CRUD and status visibility is planned for Phase 3 (Volume 4, Chapter 1.11).

2.2 Product Functions (Summary)

- Authenticate an Admin or a Collector and maintain their session.
- Admin: create/edit Projects and Tasks, and assign Collectors to them.
- Collector: retrieve and display the Projects and Tasks assigned to them.

- Verify device readiness before recording (permissions, storage, battery, network).
- Capture wide-angle, rear-facing video for a Task.
- Chunk completed recordings into 10-minute local files.
- Generate a complete, verifiable metadata record for every chunk and session.
- Queue, upload, and confirm delivery of chunks (and their metadata) to cloud storage, resiliently.
- Report sync/session status back to the Collector, and Project-wide status back to the Admin, at all times.

2.3 User Classes and Characteristics

Detailed in Chapter 1.5 (User Personas) and Chapter 1.6 (User Roles & Permissions). In summary, both current for the MVP: Collector (primary, field-based, mobile-first) and Admin (creates and manages Projects/Tasks/assignments, views status and metadata). A Super Admin class remains a future-phase concept.

2.4 Operating Environment

- Mobile OS: Android (primary target for MVP) and iOS (secondary target, build pipeline via cloud Mac service per Volume 0, Chapter 0.2, Section 2).
- Network: Cellular and Wi-Fi, including intermittent/offline conditions — the app must degrade gracefully, not fail, when offline.
- Cloud backend: AWS (S3 for storage; compute/auth provider finalized in Volume 4).

2.5 Design and Implementation Constraints

As fixed in Volume 0, Chapter 0.2 (Project Constitution): Flutter/Dart only, Windows-first development workflow, AWS S3 as the storage system of record, and offline-first architecture as a non-negotiable principle.

2.6 Assumptions and Dependencies

This SRS depends on the assumptions logged in Chapter 1.1, Section 6, and will be updated if those assumptions are resolved differently than expected.

3. System Features

Each feature below follows the pattern: description, priority, and the primary stimulus/response behavior. Full atomic requirements for each feature are enumerated in Chapter 1.3 — Functional Requirements.

3.1 Authentication & Session Management

Description: Allows a Collector to log in via email/password or enterprise SSO, and keeps them signed in without interruption via silent token refresh.

Priority: Must Have.

Stimulus/Response: Collector submits credentials → system validates and returns a session token → token is silently refreshed before expiry → on expiry without a valid refresh, the Collector is forced to re-authenticate.

3.2 Project & Task Retrieval

Description: Retrieves and displays the Projects and Tasks assigned to the authenticated Collector, with details, instructions, and requirements per Task.

Priority: Must Have.

Stimulus/Response: Collector opens Home Dashboard or Projects list → system fetches (or serves cached) Project/Task data → Collector drills into a Task to view instructions before recording.

3.3 Pre-Recording Device Health Checklist

Description: Confirms permissions, available storage, battery level, and network state before allowing recording to start.

Priority: Must Have.

Stimulus/Response: Collector proceeds from Task Detail toward Recording → system runs the checklist → any failed check blocks recording with a clear explanation and remedy.

3.4 Video Recording

Description: Captures continuous, high-bitrate, wide-angle (0.5x–0.6x) rear-facing video for the duration of the session.

Priority: Must Have.

Stimulus/Response: Collector taps Record → full-screen preview and capture begin with a visible recording indicator and elapsed timer → Collector taps Stop → capture ends and raw footage is safely written to local storage.

3.5 Automatic Chunking

Description: Splits a finished recording into fixed 10-minute chunks, performed only after Stop is tapped.

Priority: Must Have.

Stimulus/Response: Recording stops → system chunks the raw footage locally → each chunk is persisted independently before any upload begins.

3.6 Chunk Upload (Automatic / Manual)

Description: Uploads chunks to AWS S3 in the background (default) or on manual trigger (if configured), resiliently across app restarts and network loss.

Priority: Must Have.

Stimulus/Response: Chunk finalized locally → if automatic mode, upload starts immediately; if manual mode, upload starts when the Collector taps Upload → on network loss, the chunk remains queued and resumes automatically once connectivity returns.

3.7 Upload Status & Session Lifecycle

Description: Surfaces per-chunk and per-session sync status, and marks a session Complete once every chunk is confirmed uploaded.

Priority: Must Have.

Stimulus/Response: Chunk state changes (queued/uploading/failed/complete) → dashboard and session views update in real time → once all chunks in a session reach Complete, the session itself is marked Complete.

3.8 Admin Project & Task Management (New — MVP)

Description: Allows an Admin to create and edit Projects and Tasks, assign Collectors to them, and view Session/chunk status and metadata across their Projects.

Priority: Must Have.

Stimulus/Response: Admin creates/edits a Project or Task → system persists it and makes it visible only to assigned Collectors → Admin assigns a Collector → that Collector's Task list updates to include it → Admin views status/metadata for any Session under their Projects at any time.

3.9 Metadata Generation & Integrity (New — MVP)

Description: Generates a complete, immutable metadata record for every chunk at creation time, verifies its integrity end-to-end, and makes it available to Admin views.

Priority: Must Have.

Stimulus/Response: Chunk is created → system generates its full metadata record (identity, timing, technical, device/context, file integrity checksum) and persists it locally alongside the chunk → metadata uploads with (or immediately after) the chunk → backend verifies the checksum → chunk is marked Complete only once both the file and its metadata are confirmed.

4. External Interface Requirements

4.1 User Interfaces

Specified in full in Volume 2 — Product Design. This SRS only requires that every screen in the flow defined in Chapter 1.1, Section 3 exists and is reachable.

4.2 Hardware Interfaces

- Device rear-facing camera, with wide/ultra-wide lens selection (0.5x/0.6x).
- Device microphone.
- Device local storage (for chunk persistence).
- Device network radios (Wi-Fi/cellular) for upload.

4.3 Software Interfaces

- Authentication provider (credentials and/or SSO — finalized in Volume 4.7).
- Backend API for Projects/Tasks/Sessions/Metadata (finalized in Volume 4.6).
- AWS S3 for chunk storage, via resumable/multipart upload.
- Backend metadata store, associated with each chunk's S3 object key (finalized in Volume 4.4/4.5).
- Push notification service, for upload-failure and new-assignment alerts.

4.4 Communications Interfaces

All network communication uses TLS-encrypted HTTPS. Upload traffic must tolerate interruption and resume without data corruption or duplication (see Volume 5.10 — Upload Pipeline).

5. Traceability

PRD Feature	SRS System Feature
Secure login (credentials + SSO)	3.1 Authentication & Session Management
Project + Task assignment retrieval	3.2 Project & Task Retrieval
Pre-recording device health checklist	3.3 Pre-Recording Device Health Checklist
Wide-angle rear-facing recording	3.4 Video Recording
Automatic 10-minute chunking	3.5 Automatic Chunking
Automatic/manual chunk upload	3.6 Chunk Upload (Automatic / Manual)
Upload status visibility, session lifecycle	3.7 Upload Status & Session Lifecycle
Admin creates/edits Projects & Tasks, assigns Collectors	3.8 Admin Project & Task Management
Per-chunk metadata capture, integrity, and storage	3.9 Metadata Generation & Integrity

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CHAPTER 1.3

Functional Requirements

Atomic, Testable Requirements by Module

Document Title	Functional Requirements
Chapter Reference	1.3
Version	1.1
Status	Draft — Pending Approval
Owner	Faisal — Human Archive
Prepared With	Claude (Cowork)
Date	05 August 2026
Revision Note	v1.1 — Added FR-ADM (Admin Project & Task Management) and expanded metadata into its own FR-META section, per owner request.
Priority Key	Must Have / Should Have / Could Have (MoSCoW)

Each requirement below is atomic and independently testable. IDs are stable identifiers — later volumes (design, architecture, testing) reference these IDs directly rather than restating the requirement.

1. Authentication & Session (FR-AUTH)

ID	Requirement	Priority
FR-AUTH-01	The system shall allow login via email and password, for both the Admin and Collector roles.	Must Have
FR-AUTH-02	The system shall support enterprise SSO (SAML/OAuth) when enabled for a client organization.	Must Have
FR-AUTH-03	The system shall silently refresh the session token before expiry without interrupting the user.	Must Have
FR-AUTH-04	The system shall force re-authentication only when the session token cannot be refreshed.	Must Have
FR-AUTH-05	The system shall securely store authentication tokens on-device (see Volume 6.7 — Secure Storage).	Must Have
FR-AUTH-06	The system shall determine which experience (Admin or Collector) to present based on the authenticated user's role.	Must Have

2. Onboarding & Permissions (FR-ONB)

ID	Requirement	Priority
FR-ONB-01	The system shall request Camera, Microphone, Location (When In Use), Notifications, and Files access during first launch, for the Collector experience.	Must Have
FR-ONB-02	The system shall show a blocking explainer screen with a settings deep link if a required permission is denied.	Must Have
FR-ONB-03	The system shall prevent recording from starting unless both Camera and Microphone permissions are granted.	Must Have

3. Admin — Project & Task Management (FR-ADM)

New in this revision: Admin capability is required for the MVP, not a future phase. These requirements are intentionally platform-agnostic — Volume 2 (Product Design) and Volume 6 (Mobile App Architecture) will determine whether the Admin experience lives inside the same app, a separate surface, or both.

ID	Requirement	Priority
FR-ADM-01	The system shall allow an Admin to create a new Project.	Must Have
FR-ADM-02	The system shall allow an Admin to create, edit, and remove Tasks within a Project, including instructions, reference examples, and requirements.	Must Have
FR-ADM-03	The system shall allow an Admin to assign one or more Collectors to a Project and to specific Tasks within it.	Must Have
FR-ADM-04	The system shall allow an Admin to reassign or remove a Collector from a Task or Project.	Should Have
FR-ADM-05	The system shall allow an Admin to view Session and chunk status for every Task under Projects they manage.	Must Have
FR-ADM-06	The system shall allow an Admin to view (and export, where feasible) the full metadata record for any chunk or session under their Projects.	Must Have

ID	Requirement	Priority
FR-ADM-07	The system shall prevent a Collector from creating, editing, or deleting Projects or Tasks, or from assigning Collectors.	Must Have
FR-ADM-08	The system shall make a newly created or edited Project/Task visible only to the Collectors explicitly assigned to it.	Must Have

4. Collector Dashboard, Projects & Tasks (FR-PT)

ID	Requirement	Priority
FR-PT-01	The system shall display a Home Dashboard showing active projects, in-progress sessions, total recorded time, and sync status.	Must Have
FR-PT-02	The system shall display pending, uploading, and completed chunk counts on the dashboard.	Must Have
FR-PT-03	The system shall display the list of Projects assigned to the logged-in Collector.	Must Have
FR-PT-04	The system shall display the list of Tasks within a selected Project.	Must Have
FR-PT-05	The system shall display Task Detail including instructions, reference examples, and requirements.	Must Have
FR-PT-06	The system shall cache Project/Task data locally for offline browsing.	Should Have
FR-PT-07	The system shall support a Collector being assigned to more than one Project at a time.	Should Have

5. Pre-Recording Checklist (FR-CHK)

ID	Requirement	Priority
FR-CHK-01	The system shall verify Camera and Microphone permissions before allowing recording to start.	Must Have
FR-CHK-02	The system shall verify sufficient free local storage exists for at least one full chunk before recording starts.	Must Have
FR-CHK-03	The system shall verify the device battery level meets a configured minimum threshold before recording starts.	Must Have
FR-CHK-04	The system shall check network availability and inform the Collector whether upload will start immediately or be queued.	Must Have
FR-CHK-05	The system shall block recording and present a clear remedy message if any checklist item fails.	Must Have

6. Recording (FR-REC)

ID	Requirement	Priority
FR-REC-01	The system shall capture video using the rear-facing camera only.	Must Have
FR-REC-02	The system shall record using a wide-angle field of view at 0.5x or 0.6x zoom factor (Android wide lens / iPhone 11+ ultra-wide).	Must Have
FR-REC-03	The system shall display a full-screen live camera preview during recording.	Must Have
FR-REC-04	The system shall display a Record/Stop control, an elapsed-time counter, and a recording indicator.	Must Have
FR-REC-05	The system shall record continuously at a high, configured bitrate for the duration of the session.	Must Have
FR-REC-06	The system shall safely persist the raw recording to local storage immediately when Stop is tapped, before any other processing begins.	Must Have

7. Chunking & Local Processing (FR-CHNK)

ID	Requirement	Priority
FR-CHNK-01	The system shall split a completed recording into fixed 10-minute chunks.	Must Have
FR-CHNK-02	The system shall perform chunking only after recording has stopped, never during active recording.	Must Have
FR-CHNK-03	The system shall persist each chunk to local storage independently and immediately after it is produced.	Must Have
FR-CHNK-04	The system shall trigger metadata generation (Section 8 — FR-META) for each chunk immediately as part of local processing.	Must Have
FR-CHNK-05	The system shall name chunk files using a deterministic, collision-proof naming strategy (Volume 5.14).	Must Have

8. Metadata Generation & Integrity (FR-META)

New in this revision: metadata is elevated from a line item under chunking to a full, first-class requirement area. Every chunk and session must be fully described and verifiable, from creation through long-term storage.

8.1 What Must Be Captured

ID	Metadata Field(s)	Description
FR-META-01	Identity	chunk ID, session ID, project ID, task ID, collector (user) ID, device ID.
FR-META-02	Timing	sequence index within the session, start timestamp (UTC), end timestamp (UTC), duration.
FR-META-03	Technical capture detail	resolution, frame rate, bitrate, codec, zoom factor (0.5x/0.6x), camera used (rear-wide).
FR-META-04	Device & app context	device model, OS version, app version.
FR-META-05	Capture conditions	GPS location (if permitted), battery level at capture, network type at capture/upload.
FR-META-06	File integrity	file size in bytes, integrity checksum (e.g. SHA-256).
FR-META-07	Collector-authored (optional)	session notes/tags — the only fields a Collector may edit (links to FR-SEC-01).

8.2 How It Must Be Handled

ID	Requirement	Priority
FR-META-08	The system shall generate the full metadata record for a chunk at the moment the chunk itself is created — not later, and not lazily on upload.	Must Have
FR-META-09	The system shall persist chunk metadata to local storage immediately, alongside the chunk file, so it survives an app crash or force-close exactly as the video does.	Must Have
FR-META-10	The system shall compute and store an integrity checksum for every chunk at creation time.	Must Have
FR-META-11	The system shall upload each chunk's metadata to the backend together with, or immediately following, the chunk itself, associated with that chunk's S3 object key.	Must Have
FR-META-12	The system shall verify the uploaded chunk's checksum against the locally recorded checksum before marking that chunk Complete.	Must Have
FR-META-13	The system shall never allow a Collector to edit or delete system-generated metadata fields; only notes/tags are Collector-editable.	Must Have
FR-META-14	The system shall retain full metadata for a session per the retention policy (Volume 8.7), even after local chunk files are cleaned up (Volume 5.15).	Must Have
FR-META-15	The system shall make session/chunk metadata available to Admin views (FR-ADM-06).	Must Have

9. Upload (FR-UPL)

ID	Requirement	Priority
FR-UPL-01	The system shall upload chunks automatically in the background by default.	Must Have
FR-UPL-02	The system shall support a manual upload mode, configurable per device, where the Collector taps Upload to begin.	Must Have
FR-UPL-03	The system shall upload chunks directly to AWS S3 using resumable/multipart upload.	Must Have
FR-UPL-04	The system shall allow the Collector to start a new recording while a previous session's chunks are still uploading.	Must Have
FR-UPL-05	The system shall queue chunks locally when offline and resume upload automatically once network connectivity returns.	Must Have
FR-UPL-06	The system shall retry failed uploads automatically using a defined backoff strategy (Volume 5.13 — Retry Strategy).	Must Have
FR-UPL-07	The system shall allow the Collector to manually retry a failed chunk upload.	Should Have
FR-UPL-08	The system shall allow the Collector to pause and resume an in-progress upload.	Should Have
FR-UPL-09	The system shall never produce a duplicated or corrupted chunk in S3 as a result of a retried or resumed upload.	Must Have

10. Session Lifecycle & Status (FR-SES)

ID	Requirement	Priority
FR-SES-01	The system shall track the state of every session (recording, chunking, uploading, complete) end to end.	Must Have
FR-SES-02	The system shall mark a session Complete only once every one of its chunks — and their metadata — is confirmed uploaded (FR-META-12).	Must Have
FR-SES-03	The system shall display per-session and per-chunk upload/sync status to the Collector, and Project-wide status to the Admin.	Must Have
FR-SES-04	The system shall return the Collector to the Task List or Dashboard once a session is marked Complete.	Must Have

11. Secondary Features (FR-SEC)

ID	Requirement	Priority
FR-SEC-01	The system shall allow the Collector to add notes/tags to a recording session.	Should Have
FR-SEC-02	The system shall display reference media (images/video) attached to a Task's instructions.	Should Have
FR-SEC-03	The system shall send a push notification when a chunk upload fails.	Should Have
FR-SEC-04	The system shall send a push notification when a new Task is assigned to the Collector.	Should Have
FR-SEC-05	The system shall provide an in-app support/help contact option.	Could Have

VOLUME 1 — PRODUCT PLANNING

CHAPTER 1.4

Non-Functional Requirements

How Well the System Must Perform

Document Title	Non-Functional Requirements
Chapter Reference	1.4
Version	1.1
Status	Draft — Pending Approval
Owner	Faisal — Human Archive
Prepared With	Claude (Cowork)
Date	05 August 2026
Revision Note	v1.1 — Added Section 11 (Metadata Integrity) to match the elevation of metadata to a first-class requirement in Chapter 1.3.
Companion Document	Chapter 1.3 — Functional Requirements

Targets below are initial engineering targets to guide design; they will be tightened or relaxed once a pilot cohort of Collectors provides real usage data (Volume 11 — Project Management).

1. Performance

ID	Requirement	Target / Measure
NFR-PERF-01	Recording preview shall start within a short, bounded time of the Collector opening the Recording Screen.	< 1.5 seconds on a mid-tier device
NFR-PERF-02	Chunking of a completed recording shall begin immediately after Stop is tapped.	< 2 seconds to begin local chunking
NFR-PERF-03	The app shall remain responsive (UI thread unblocked) while chunking and uploading run in the background.	No dropped frames / ANRs during upload

2. Reliability & Resilience

ID	Requirement	Target / Measure
NFR-REL-01	No recorded footage shall be lost due to an app crash after Stop has been tapped.	Zero data-loss incidents per Volume 0, Chapter 0.1, Section 4
NFR-REL-02	Chunk uploads shall be resumable after any interruption (app kill, network loss, device restart).	100% of interrupted uploads resume without restarting from zero
NFR-REL-03	Retried or resumed uploads shall never create duplicate or corrupted chunks in S3.	Verified via idempotency checks — see Volume 5.10
NFR-REL-04	The app shall recover its full local queue state after a force-close or crash.	Queue state fully restored on next launch

3. Availability

ID	Requirement	Target / Measure
NFR-AVL-01	Core recording functionality (viewing tasks, recording, local chunking) shall be available with zero network connectivity.	100% offline availability for these workflows
NFR-AVL-02	Upload shall resume automatically within a short window of connectivity being restored, with no manual action required.	< 30 seconds after network restoration

4. Scalability

ID	Requirement	Target / Measure
NFR-SCL-01	The upload queue shall handle many chunks queued simultaneously without degrading device performance.	Queue depth of 50+ pending chunks without UI slowdown
NFR-SCL-02	The backend and storage layer shall scale to a growing fleet of concurrent Collectors uploading simultaneously.	Verified under load during Volume 9 — Testing

5. Security & Privacy

ID	Requirement	Target / Measure
NFR-SEC-01	All network communication shall be encrypted in transit (TLS).	TLS 1.2+ enforced on all endpoints
NFR-SEC-02	Authentication tokens shall be stored using platform secure storage, never in plain text.	Verified in Volume 6.7 and Volume 8.2

ID	Requirement	Target / Measure
NFR-SEC-03	Locally stored video chunks shall be treated as sensitive until upload is confirmed.	Access restricted to the app sandbox; encryption evaluated in Volume 8.2
NFR-SEC-04	The system shall comply with the data retention policy defined in Volume 8.7.	Reviewed against applicable privacy regulations

6. Usability

ID	Requirement	Target / Measure
NFR-USE-01	A new Collector shall be able to complete onboarding and produce their first successful recording without external help.	Time-to-first-recording tracked per Volume 0, Chapter 0.1, Section 4
NFR-USE-02	Recording and upload status shall be understandable at a glance, without needing to open a detail screen.	Validated via usability review in Volume 2 — Product Design
NFR-USE-03	Error and permission-denial messages shall clearly state what happened and what the Collector must do next.	Reviewed against UX guidelines in Volume 2.9

7. Maintainability

ID	Requirement	Target / Measure
NFR-MNT-01	The codebase shall follow the layered architecture and coding standards fixed in Volume 0, Chapter 0.2, Sections 3–4.	Verified via code review checklist, Volume 12.5
NFR-MNT-02	Capture parameters (chunk length, resolution, bitrate) shall be adjustable without a full app-store release once remote config ships.	Delivered per Volume 3.3 / Chapter 11 roadmap

8. Portability

ID	Requirement	Target / Measure
NFR-PRT-01	The application shall run on both Android and iOS from a single Flutter codebase.	Verified per Volume 6 — Mobile App Architecture
NFR-PRT-02	Platform-specific camera behavior (Android wide lens vs. iPhone ultra-wide) shall be abstracted behind a common recording interface.	Verified in Volume 5.1 — Camera Module

9. Resource Efficiency

ID	Requirement	Target / Measure
NFR-RES-01	Recording and background upload shall not cause excessive battery drain during a typical work shift.	Benchmarked on target devices during Volume 9 — Testing
NFR-RES-02	Local storage cleanup shall automatically remove chunks after upload is confirmed, per configured retention rules.	Verified against Volume 5.15 — Storage Cleanup

10. Localization Readiness

ID	Requirement	Target / Measure
NFR-LOC-01	All user-facing strings shall be externalized to support future multi-language localization.	No hard-coded UI strings by MVP code-freeze

11. Metadata Integrity (New)

ID	Requirement	Target / Measure
NFR-META-01	Metadata generation for a chunk shall not measurably delay chunk finalization.	< 500ms added overhead per chunk
NFR-META-02	Checksum verification shall catch corrupted or truncated chunk uploads before they are marked Complete.	100% of corrupted/truncated uploads caught pre-Complete
NFR-META-03	No chunk shall ever be marked Complete without a fully populated, backend-confirmed metadata record.	Zero exceptions — enforced server-side, not just client-side
NFR-META-04	Metadata shall remain queryable by Admins even after the underlying local chunk file has been cleaned up.	100% metadata availability through the retention window (Volume 8.7)

VOLUME 1 — PRODUCT PLANNING

CHAPTER 1.5

User Personas

Who Uses This Product, and Why

Document Title	User Personas
Chapter Reference	1.5
Version	1.1
Status	Draft — Pending Approval
Owner	Faisal — Human Archive
Prepared With	Claude (Cowork)
Date	05 August 2026
Revision Note	v1.1 — "Aman" (Project/Operations Admin) is promoted from a future-phase persona to a current, MVP-scope persona, per owner request.
Basis	Roles implied by the confirmed product flow and feature list (Chapter 1.1)

Three personas are defined: two current, MVP-scope personas (the Collector and the Admin), and one persona tied to a later phase (Enterprise IT Admin) that shapes decisions made now even though its interface ships later.

1. Primary Persona — The Field Collector (Current — MVP)

"Riya" — Field Collector

Records assigned footage in the field, often alone, often with unreliable connectivity.

Role: Primary end user of the mobile app; paid or contracted to record video against specific tasks.

Context: Works on-site or on-the-move across multiple locations in a single shift; may go hours without reliable Wi-Fi or cellular data. Often manages several assigned tasks across one or more projects in a single day.

Primary Device: A mid-range Android phone (primary), occasionally an iPhone depending on client-provided hardware.

Technical Comfort: Comfortable with everyday consumer apps; not a technical user. Needs clear, unambiguous guidance rather than technical error messages.

Goals

- Understand exactly what to record before starting, without needing to ask anyone.
- Record without worrying about losing footage or running out of storage mid-session.
- Keep working (recording the next task) without waiting for the previous recording to finish uploading.
- Know, at a glance, whether their work has actually made it to the server.

Pain Points Today

- Today, recording apps aren't built for structured tasks — there's no link between a recording and the task it belongs to.
- Long recordings that fail or get lost represent hours of unpaid, wasted effort.
- No visibility into whether an upload actually succeeded, especially on bad networks.
- Being blocked from starting new work because the app is busy uploading the last session.

What Success Looks Like

- Every session recorded is fully accounted for — chunked, queued, and eventually confirmed uploaded — without Riya doing anything manual.
- Riya can start a new recording immediately after stopping the previous one.
- When something does fail, Riya sees a clear status and a simple retry action.

2. Secondary Persona (Current — MVP) — Project / Operations Admin

"Aman" — Project / Operations Admin

Creates projects and tasks, assigns collectors, and needs to trust the metadata behind every session.

Role: Creates and manages Projects and Tasks, assigns Collectors, and reviews Session/chunk status and metadata. Now a current, MVP-scope role, not a future one.

Context: Manages one or more concurrent Projects and a set of Collectors, some in-house and some contracted. May work from a mobile screen inside the same app, a lightweight web view, or both — the exact surface is an open question for Volume 2/Volume 6.

Primary Device: Desktop/laptop browser and/or the mobile app, depending on what Volume 2 (Product Design) decides.

Technical Comfort: Comfortable with dashboards, spreadsheets, and basic reporting tools; not a developer.

Goals

- Stand up a new Project and its Tasks quickly, with clear instructions Collectors can follow unsupervised.
- Assign the right Collectors to the right Tasks without needing engineering help.
- See collection progress and status across active Projects and Collectors.
- Trust that the metadata behind every chunk and session is complete and hasn't been tampered with.

Pain Points Today

- No current tool ties task assignment to actual footage delivered.
- No visibility into upload failures until a Collector reports them manually.
- No way today to verify that footage tagged to a Task is actually what it claims to be (right project, right time, right device) without watching it end to end.

What Success Looks Like

- A Project and its Tasks can be created and assigned in minutes, without developer involvement.
 - A clear view of collection status across all active Projects and Collectors.
 - Confidence that a Task marked "complete" actually has usable, fully-described footage behind it — because the metadata proves it.
-

3. Tertiary Persona (Future Phase) — Enterprise IT / Client Admin

"Priya" — Enterprise IT / Client Admin

Configures how her organization's Collectors authenticate and what data-handling rules apply.

Role: IT or compliance stakeholder at a client organization that uses the platform for its own data-collection program.

Context: Not a daily user of the app; engages primarily during onboarding/configuration and periodic compliance reviews.

Primary Device: Desktop/laptop, primarily for configuration, not daily operations.

Technical Comfort: Technical, security- and compliance-conscious; cares about SSO integration and data retention more than day-to-day workflow.

Goals

- Enforce enterprise SSO (SAML/OAuth) for all Collectors under her organization instead of standalone passwords.
- Confirm data retention and privacy handling meet her organization's compliance requirements (Volume 8.6, 8.7).
- Have a single point of contact/process for security questions or incidents.

Pain Points Today

- Vendor tools that don't support enterprise identity providers create shadow accounts and audit gaps.
- Unclear data retention policies are a compliance risk.

What Success Looks Like

- SSO is enforced with no ability for Collectors under her org to bypass it.
- A clear, written data retention and privacy policy she can hand to her own compliance team.

VOLUME 1 — PRODUCT PLANNING

CHAPTER 1.6

User Roles & Permissions

Who Can Do What

Document Title	User Roles & Permissions
Chapter Reference	1.6
Version	1.1
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Prepared With	Claude (Cowork)
Date	05 August 2026
Revision Note	v1.1 — Admin is promoted from a future-phase role to a current, MVP-scope role; metadata view/export capability is added to the permission matrix, per owner request.
Companion Document	Chapter 1.5 — User Personas

1. Roles Defined

Admin (Current — MVP)

Creates and edits Projects and Tasks (the "lists" of work), assigns Collectors to them, and can view Session/chunk status and metadata for everything under their Projects. This role is now in scope for the MVP — it is no longer deferred to a future phase. The exact Admin surface (inside the same app, a separate web view, or both) is an open question tracked in Chapter 1.1, Section 8, and resolved in Volume 2/Volume 6.

Collector / User (Current — MVP)

Can only see and act on Projects/Tasks explicitly assigned to them by an Admin. Records, chunks, and uploads footage; cannot create or edit Projects/Tasks, and cannot assign work to anyone, including themselves.

Super Admin (Future Phase)

Full platform control: manages Admins themselves, client-level configuration (e.g. SSO settings, remote capture configuration), and cross-project analytics. Still a future-phase role — not required for MVP.

System (Background Service)

Not a human role — represents automated processes (chunking engine, upload engine, metadata generation, token refresh) that act on a user's behalf without direct input.

2. Permission Matrix

"✓" = permitted, "—" = not permitted, "View only" = read access without modification rights. The Admin column is now current/MVP scope; only the Super Admin column remains future-phase.

Capability	Collector	Admin (Current)	Super Admin (Future)
Log in via credentials or SSO	✓	✓	✓
View own assigned Projects/Tasks	✓	✓ (all managed)	✓ (all)
Create/edit Projects	—	✓	✓
Create/edit Tasks	—	✓	✓
Assign Collectors to Projects/Tasks	—	✓	✓
Record video for an assigned Task	✓	—	—
View own recording/upload status	✓	View only	View only
View Session/chunk status for own Projects	—	✓	✓
View & export chunk/session metadata for own Projects	View own only	✓	✓
View fleet-wide upload/collection status (all Projects)	—	—	✓
Flag a session/chunk for rework (QA) — Phase 3	—	✓	✓
View collector performance analytics — Phase 3	—	✓	✓

Capability	Collector	Admin (Current)	Super Admin (Future)
Configure remote capture parameters (chunk length, bitrate, resolution) — Phase 3	—	—	✓
Manage other Admin accounts	—	—	✓
Configure client-level SSO settings	—	—	✓

3. Notes on Enforcement

- All permission checks are enforced server-side, not just hidden in the mobile/Admin UI (Volume 4.8 — Authorization Design).
- A Collector's visibility into Projects/Tasks is scoped strictly to their own assignments — there is no "browse all projects" capability for the Collector role.
- The Admin role is in scope for the MVP: an Admin can only create/edit/assign within their own Projects, not across every client or organization on the platform.
- Super Admin remains a future-phase role, documented here so the backend data model (Volume 4) supports it from day one and avoids a breaking rework later.
- Metadata visibility follows Project ownership: a Collector sees only their own session/chunk metadata; an Admin sees metadata for every Session under Projects they manage.

VOLUME 1 — PRODUCT PLANNING

CHAPTER 1.7

User Stories

Requirements Framed From the Collector's Perspective

Document Title	User Stories
Chapter Reference	1.7
Version	1.1
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Prepared With	Claude (Cowork)
Date	05 August 2026
Revision Note	v1.1 — Added an Admin epic (project/task management, metadata visibility) now that Admin is in MVP scope, per owner request.
Format	As a <role>, I want <capability>, so that <benefit>.

Every story below traces back to one or more Functional Requirements in Chapter 1.3, so nothing here introduces new scope — it only reframes the same requirements from the user's point of view for design and testing purposes.

Epic: Onboarding & Access

ID	User Story	Linked FR(s)
US-01	As a Collector, I want to be clearly told why the app needs Camera, Microphone, Location, Notifications, and Files access, so that I trust the app and grant the right permissions.	FR-ONB-01
US-02	As a Collector, I want to be shown exactly how to fix a denied permission, so that I'm not stuck without knowing what to do.	FR-ONB-02
US-03	As a Collector, I want to log in with my work credentials or my organization's SSO, so that I don't need a separate password just for this app.	FR-AUTH-01, FR-AUTH-02
US-04	As a Collector, I want to stay logged in across sessions without repeated prompts, so that I can get straight to work each shift.	FR-AUTH-03, FR-AUTH-04

Epic: Dashboard & Task Discovery

ID	User Story	Linked FR(s)
US-05	As a Collector, I want to see all my active projects and in-progress sessions on one dashboard, so that I know what's outstanding at a glance.	FR-PT-01
US-06	As a Collector, I want to see how many chunks are pending, uploading, or completed, so that I understand my sync status without digging into detail screens.	FR-PT-02
US-07	As a Collector, I want to open a Project and see its Task list, so that I can pick the right Task to work on.	FR-PT-03, FR-PT-04
US-08	As a Collector, I want to read a Task's instructions and reference examples before recording, so that I record exactly what's required the first time.	FR-PT-05
US-09	As a Collector, I want to browse my assigned Projects and Tasks even with no signal, so that a bad network doesn't stop me from planning my work.	FR-PT-06

Epic: Pre-Recording Readiness

ID	User Story	Linked FR(s)
US-10	As a Collector, I want the app to check my permissions, storage, battery, and network before I start recording, so that I don't waste a recording on a device that isn't ready.	FR-CHK-01–04
US-11	As a Collector, I want a clear explanation and fix when a readiness check fails, so that I can resolve it quickly and get back to work.	FR-CHK-05

Epic: Recording

ID	User Story	Linked FR(s)
US-12	As a Collector, I want a full-screen wide-angle camera preview with a visible Record/Stop control and timer, so that I always know I'm capturing correctly.	FR-REC-01–04

ID	User Story	Linked FR(s)
US-13	As a Collector, I want my footage safely saved the instant I tap Stop, so that a crash right after stopping never costs me the recording.	FR-REC-06

Epic: Chunking & Upload

ID	User Story	Linked FR(s)
US-14	As a Collector, I want my finished recording automatically split into 10-minute chunks, so that one huge file isn't at risk of failing to upload as a whole.	FR-CHNK-01, FR-CHNK-02
US-15	As a Collector, I want my chunks to start uploading automatically in the background, so that I don't have to babysit the process.	FR-UPL-01, FR-UPL-03
US-16	As a Collector working in a controlled or low-bandwidth setting, I want the option to trigger uploads manually instead, so that I control when data leaves my device.	FR-UPL-02
US-17	As a Collector, I want to start recording my next Task immediately, even while the last session is still uploading, so that I'm never blocked waiting on the network.	FR-UPL-04
US-18	As a Collector working somewhere with no signal, I want my chunks to queue locally and upload automatically once I'm back online, so that I don't lose work to bad connectivity.	FR-UPL-05, FR-UPL-06
US-19	As a Collector, I want to manually retry a chunk that failed to upload, so that I have control when automatic retries aren't enough.	FR-UPL-07
US-20	As a Collector, I want to pause and resume an upload, so that I can manage my data usage on a limited mobile plan.	FR-UPL-08

Epic: Status & Completion

ID	User Story	Linked FR(s)
US-21	As a Collector, I want to see the live sync status of every chunk and session, so that I always know what's actually safe in the cloud.	FR-SES-03
US-22	As a Collector, I want a session to be clearly marked Complete once everything has uploaded, so that I know I'm truly done with that Task.	FR-SES-01, FR-SES-02
US-23	As a Collector, I want to be taken back to my Task List or Dashboard once a session completes, so that I can immediately move to my next assignment.	FR-SES-04

Epic: Secondary & Notification Features

ID	User Story	Linked FR(s)
US-24	As a Collector, I want to add a note or tag to a session, so that I can flag anything unusual about that recording for review.	FR-SEC-01
US-25	As a Collector, I want to be notified if a chunk fails to upload, so that I can act on it instead of assuming everything worked.	FR-SEC-03
US-26	As a Collector, I want to be notified when I'm assigned a new Task, so that I don't have to keep checking the app manually.	FR-SEC-04

Epic: Admin — Project & Task Management (New — MVP)

ID	User Story	Linked FR(s)
US-27	As an Admin, I want to create a new Project, so that I have somewhere to organize a client's data-collection work.	FR-ADM-01

ID	User Story	Linked FR(s)
US-28	As an Admin, I want to create and edit Tasks within a Project, including instructions and reference examples, so that Collectors know exactly what to record.	FR-ADM-02
US-29	As an Admin, I want to assign specific Collectors to a Project or Task, so that only the right people see and can act on that work.	FR-ADM-03
US-30	As an Admin, I want to reassign or remove a Collector from a Task, so that I can adapt when availability or performance changes.	FR-ADM-04
US-31	As an Admin, I want to see the Session and chunk status for every Task under my Projects, so that I always know how collection is progressing.	FR-ADM-05
US-32	As an Admin, I want to view and export the metadata behind any chunk or session in my Projects, so that I can verify the footage is genuine, complete, and correctly attributed.	FR-ADM-06
US-33	As a Collector, I want to be confident that only Admins can create, edit, or assign Projects and Tasks, so that the work I see is always legitimate and intentional.	FR-ADM-07

VOLUME 1 — PRODUCT PLANNING

CHAPTER 1.8

Use Cases

Detailed Interaction Flows, Including Alternate and Exception Paths

Document Title	Use Cases
Chapter Reference	1.8
Version	1.1
Status	Draft — Pending Approval
Owner	Faisal — Human Archive
Prepared With	Claude (Cowork)
Date	05 August 2026
Revision Note	v1.1 — Added UC-07 (Admin creates a Project and assigns a Collector), now that Admin is in MVP scope.
Companion Documents	Chapter 1.3 (Functional Requirements), Chapter 1.7 (User Stories)

These use cases expand the most important and highest-risk flows into full main/alternate/exception detail, to remove ambiguity before design (Volume 2) and engineering (Volumes 3–6) begin.

Use Cases

UC-01 — Record and Upload a Task Session (Happy Path)

Actor: Collector

Trigger: Collector selects a Task and taps Record.

Preconditions: Collector is logged in; Camera and Microphone permissions are granted; the Task is assigned to the Collector.

Main Flow:

1. Collector opens Task Detail and reviews instructions, examples, and requirements.
2. Collector proceeds to the Pre-Recording Checklist.
3. System verifies permissions, storage, battery, and network, and all checks pass.
4. Collector taps Record; the system displays a full-screen wide-angle preview with a recording indicator and timer.
5. Collector records the required footage and taps Stop.
6. System safely persists the raw recording to local storage.
7. System automatically splits the recording into 10-minute chunks and generates metadata for each.
8. System begins uploading chunks automatically in the background.
9. System updates per-chunk and per-session status as each chunk is confirmed uploaded.
10. Once all chunks are confirmed, the system marks the session Complete and returns the Collector to the Task List.

Postconditions: All chunks for the session are present in S3; the session is marked Complete; the Collector is free to start another recording.

UC-02 — Resume Upload After Network Loss

Actor: Collector

Trigger: Network connectivity drops while one or more chunks are uploading.

Preconditions: A recording session has been chunked and at least one chunk is queued or mid-upload.

Main Flow:

1. System detects the loss of network connectivity.
2. System pauses in-progress uploads and marks affected chunks as queued.

3. Collector continues using the app normally (e.g. starts a new recording) while offline.
4. Network connectivity is restored.
5. System automatically resumes uploading queued/interrupted chunks without any action from the Collector.
6. System updates upload status as each chunk completes.

Alternate Flow:

- If a chunk was partially uploaded before the interruption, the system resumes from the last confirmed byte range rather than re-uploading the entire chunk (see Volume 5.10 — Upload Pipeline).

Exception Flow:

- If network remains unavailable for an extended period, chunks remain safely queued locally indefinitely, and the Collector can continue recording new sessions without restriction.

Postconditions: All previously queued chunks are confirmed uploaded once connectivity allows, with no duplication or data loss.

UC-03 — Login via Enterprise SSO

Actor: Collector (belonging to a client organization with SSO enabled)

Trigger: Collector opens the app and selects "Sign in with SSO."

Preconditions: The Collector's organization has SSO (SAML/OAuth) configured for this platform.

Main Flow:

1. Collector selects the SSO login option.
2. App redirects to the organization's identity provider.
3. Collector authenticates with their organizational credentials.
4. Identity provider returns an authentication token to the app.
5. System exchanges the token for an app session and signs the Collector in.
6. System silently refreshes the session token in the background going forward.

Exception Flow:

- If the identity provider rejects the credentials, the app displays the provider's error and allows the Collector to retry.
- If the token exchange fails, the app displays a clear error and offers a retry option rather than a generic failure.

Postconditions: Collector is signed in and can access their assigned Projects and Tasks.

UC-04 — Pre-Recording Health Check Failure

Actor: Collector

Trigger: Collector attempts to proceed from Task Detail to the Recording Screen.

Preconditions: Collector is logged in and has selected a Task.

Main Flow:

1. System runs the Pre-Recording Checklist (permissions, storage, battery, network).
2. One or more checks fail (e.g. battery below threshold).
3. System blocks entry to the Recording Screen and displays which check(s) failed and why.
4. System offers a clear remedy (e.g. "Charge your device to at least 20% before recording").
5. Collector resolves the issue and re-attempts the checklist.
6. All checks pass and the Collector proceeds to Recording.

Postconditions: Recording is never started on a device that fails the readiness checklist.

UC-05 — Start a New Recording While a Previous Session Uploads

Actor: Collector

Trigger: Collector taps Record for a new Task while a previous session's chunks are still uploading.

Preconditions: A previous session exists in an uploading (not yet Complete) state.

Main Flow:

1. Collector navigates from the Dashboard/Task List to a new Task.
2. System runs the Pre-Recording Checklist for the new session independently of the previous session's upload state.
3. Collector records and stops the new session as normal (per UC-01).
4. System chunks and queues the new session's chunks for upload alongside the still-uploading previous session.
5. Both sessions' uploads progress concurrently, each tracked independently on the dashboard.

Postconditions: Both sessions eventually reach Complete independently; recording is never blocked by an in-progress upload.

UC-06 — Manually Retry a Failed Chunk Upload

Actor: Collector

Trigger: A chunk's upload has failed after exhausting automatic retries.

Preconditions: At least one chunk is in a Failed state, visible on the dashboard or session detail screen.

Main Flow:

1. Collector opens the session/chunk detail view and sees the Failed status.
2. Collector taps Retry on the failed chunk.

3. System re-attempts the upload using the same resumable upload mechanism.
4. Upload succeeds and the chunk's status updates to Complete.

Exception Flow:

- If the manual retry also fails, the system keeps the chunk in a Failed state, preserves the local file, and surfaces the failure again rather than silently dropping it.

Postconditions: The chunk either reaches Complete or remains visibly Failed and locally preserved for a future retry — it is never silently lost.

UC-07 — Admin Creates a Project, Adds a Task, and Assigns a Collector

Actor: Admin

Trigger: Admin needs to stand up a new piece of data-collection work.

Preconditions: Admin is logged in.

Main Flow:

1. Admin creates a new Project, giving it a name and description.
2. Admin creates a Task within the Project, including instructions, reference examples, and requirements.
3. Admin selects one or more Collectors and assigns them to the Task.
4. System makes the Project/Task visible only to the assigned Collector(s), in their Task List.
5. Admin can, at any point afterward, view the Session/chunk status and metadata for that Task.

Alternate Flow:

- Admin edits an existing Task's instructions instead of creating a new one — the update is immediately visible to assigned Collectors.
- Admin reassigns a Task from one Collector to another; the Task disappears from the first Collector's list and appears in the second's.

Exception Flow:

- If the Admin attempts to assign a Collector who does not have an account or is deactivated, the system blocks the assignment and explains why.

Postconditions: The Project/Task exists, is visible only to its assigned Collector(s), and the Admin can track its status and metadata going forward.

VOLUME 1 — PRODUCT PLANNING**CHAPTER 1.9**

Business Rules

Constraints the System Must Always Enforce

Document Title	Business Rules
Chapter Reference	1.9
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Status	Draft — Pending Approval
Owner	Faisal — Human Archive
Prepared With	Claude (Cowork)
Date	05 August 2026
Revision Note	v1.1 — Added Section 7 (Admin & Assignment Rules) and Section 8 (Metadata Integrity Rules), per owner request.
Nature	These are policy constraints, not features — they apply regardless of which screen or flow triggers them.

1. Recording Rules

ID	Rule	Rationale
BR-01	Recording shall only use the rear-facing camera; the front camera is never used for task recording in the MVP.	Egocentric capture requires a consistent, forward-facing viewpoint.
BR-02	Recording shall only be performed at a wide-angle zoom factor of 0.5x or 0.6x — no other zoom level is permitted for task footage.	Consistent field of view across all Collectors is required for usable training data.
BR-03	Recording shall not be permitted unless both Camera and Microphone permissions are granted.	The product requires audio-visual capture; partial capture is not a valid product state.
BR-04	Recording shall not be permitted unless the Pre-Recording Checklist passes (permissions, storage, battery, network).	Prevents starting a session that is likely to fail or be lost.

2. Chunking Rules

ID	Rule	Rationale
BR-05	Chunking shall occur only after the Collector taps Stop — never during active recording.	Splitting mid-recording risks corrupting the active capture stream.
BR-06	Each chunk shall be exactly 10 minutes in length, except for the final chunk of a session, which may be shorter.	Fixed-length chunks bound both upload size and worst-case data loss per chunk.
BR-07	Every chunk shall be persisted to local storage before any upload of that chunk begins.	Guarantees a local copy survives even if upload never starts (e.g. permanently offline).

3. Upload Rules

ID	Rule	Rationale
BR-08	A chunk shall not be deleted from local storage until its upload to S3 is confirmed.	Prevents data loss from a premature or failed cleanup.
BR-09	Automatic upload is the default mode; manual upload is only used when explicitly configured for a device.	Minimizes Collector burden while still allowing controlled environments to opt out.
BR-10	The Collector shall be able to start a new recording at any time, regardless of the upload state of any other session.	Recording throughput must never be gated on upload speed or network quality.
BR-11	A retried or resumed upload shall never create a duplicate object in S3 for the same chunk.	Dataset integrity depends on exactly-once delivery per chunk.

4. Session Lifecycle Rules

ID	Rule	Rationale
BR-12	A session shall be marked Complete only when every one of its chunks is confirmed uploaded.	"Complete" must mean the data is actually safe in cloud storage, not just recorded.
BR-13	A session that is interrupted (e.g. app killed mid-recording) shall preserve all footage recorded up to the last safely written point.	Partial data is more valuable than no data, and must never be silently discarded.

5. Access & Assignment Rules

ID	Rule	Rationale
BR-14	A Collector shall only see Projects and Tasks explicitly assigned to them.	Prevents unauthorized visibility into other clients' or collectors' work.
BR-15	A Collector may be assigned to more than one Project concurrently.	Reflects real-world usage where Collectors juggle multiple engagements.

6. Configuration Rules

ID	Rule	Rationale
BR-16	Chunk duration, resolution, and bitrate are configuration values, not hard-coded constants, even before remote configuration ships.	Enables Volume 3.3 remote configuration without a later architectural rework.
BR-17	Upload mode (automatic vs. manual) is configurable per device, not globally fixed.	Different deployment contexts (e.g. metered vs. unmetered networks) require different defaults.

7. Admin & Assignment Rules (New)

ID	Rule	Rationale
BR-18	Only an Admin may create, edit, or delete Projects and Tasks, or assign Collectors to them.	Keeps the structure of work under deliberate control, not ad hoc collector self-service.
BR-19	A Collector shall never have visibility into a Project or Task they are not assigned to, regardless of Admin-side changes made elsewhere in real time.	Prevents cross-client or cross-collector data leakage as Projects scale up.
BR-20	An Admin may only manage Projects, Tasks, and Collectors within their own scope — not across every organization on the platform.	Multi-tenant safety; full cross-organization control is reserved for the future Super Admin role.

8. Metadata Integrity Rules (New)

ID	Rule	Rationale
BR-21	A chunk is not considered valid or Complete unless its full metadata record has also been confirmed received by the backend.	"Complete" must mean the footage is both present and fully described/verifiable, not just uploaded.
BR-22	System-generated metadata fields are immutable once written; only Collector-authored notes/tags may ever be edited after the fact.	Metadata must remain trustworthy evidence of what was actually captured, when, and by whom.
BR-23	Metadata for a session shall be retained per the data retention policy (Volume 8.7) even after the underlying local chunk file has been cleaned up.	Admins and future QA workflows must be able to audit a session long after the raw file is gone.

VOLUME 1 — PRODUCT PLANNING

CHAPTER 1.10

MVP Scope

What Ships First, and What Explicitly Does Not

Document Title	MVP Scope
Chapter Reference	1.10
Version	1.1
Status	Draft — Pending Approval
Owner	Faisal — Human Archive
Prepared With	Claude (Cowork)
Date	05 August 2026
Revision Note	v1.1 — Core Admin capability (create/edit Projects & Tasks, assign Collectors, view status/metadata) moves into MVP scope; full metadata capture/integrity is now an explicit MVP requirement. Only the advanced Admin/Ops tooling (analytics, QA/rework, remote config, fleet-wide monitoring) stays deferred to Phase 3.
Purpose	Prevent scope creep during Volumes 3–9 by fixing what "MVP done" means.

1. MVP Definition

The MVP is the smallest version of the product that lets an Admin structure the work (Projects, Tasks, Collector assignment) and lets a Collector complete the full confirmed flow — install, onboard, log in, view assigned work, record, chunk, upload, and see completion — reliably, without data loss, and with complete metadata behind every chunk, on Android first.

2. In Scope for MVP

2.1 Collector Capabilities

- Secure login (credentials + SSO) with silent token refresh.
- Project + Task assignment retrieval, including Task Detail (instructions, examples, requirements).
- Pre-recording device health checklist (permissions, storage, battery, network).
- Rear-facing, wide-angle (0.5x/0.6x) continuous, high-bitrate video recording.
- Automatic 10-minute chunking after Stop (never during recording).
- Immediate local persistence of each chunk.
- Automatic upload by default, with a manual-upload mode configurable per device.
- Background, resumable, chunked upload directly to S3.
- Offline recording with local queueing and automatic retry on reconnect.
- Full session lifecycle tracking through to "Complete."
- Upload status visibility at both dashboard and per-session level.

2.2 Admin Capabilities (New — Promoted from Future Phase)

- Create, edit, and remove Projects.
- Create, edit, and remove Tasks within a Project, including instructions, reference examples, and requirements.
- Assign one or more Collectors to a Project and/or specific Tasks.
- Reassign or remove a Collector from a Task.
- View Session/chunk status for every Task under Projects they manage.
- View and export the full metadata record for any chunk or session under their Projects.

2.3 Metadata (New — Explicit MVP Requirement)

- Full per-chunk metadata capture at creation time: identity, timing, technical capture detail, device/app context, capture conditions, and file integrity checksum (Chapter 1.3, Section 8).

- Local persistence of metadata immediately alongside its chunk, surviving crashes and restarts.
- Metadata uploaded with (or immediately after) its chunk and checksum-verified before the chunk is marked Complete.
- Metadata retained per the retention policy even after local chunk cleanup.

Platform: Android is the primary MVP target and release platform; iOS is built in parallel where feasible but is not a release blocker for MVP (Volume 0, Chapter 0.2, Section 2). The Admin surface's exact platform (mobile, web, or both) is an open question — see Chapter 1.1, Section 8.

3. Explicitly Out of Scope for MVP

These are real, committed features — just not part of the first release. Building toward them is still allowed in the architecture (see Business Rules BR-16/BR-17), but none are required to work before MVP sign-off.

3.1 Deferred to Phase 2 (Secondary Collector Features)

- Session notes/tagging.
- Reference media attached to task instructions.
- Push notifications (upload failures, new assignments).
- Multi-project support polish (basic multi-project access may exist in MVP; refined UX is Phase 2).
- Offline browsing polish/full caching strategy.
- In-app support/help contact.
- Manual pause/resume upload controls (manual retry of a fully failed chunk is in MVP; pause/resume mid-upload is Phase 2).

3.2 Deferred to Phase 3 (Advanced Admin & Ops — Core Admin Is Now MVP)

- Live, fleet-wide upload/collection monitoring dashboard (visual analytics across every Project).
- Dataset QA/review and rework-flagging workflow.
- Collector performance and throughput analytics.
- Remote configuration of chunk duration, resolution, and bitrate via feature flags.

3.3 Deferred to Phase 4 (Future Capabilities)

- OTA remote configuration without app-store releases.
- AI-assisted in-app QA (blur/occlusion/motion-blur detection).
- Multi-language localization.

- Simultaneous multi-camera capture (front + rear).
- Wearable capture device integration.

4. MVP Acceptance Criteria

The MVP is considered done when all of the following are demonstrably true on a target Android device:

#	Acceptance Criterion
1	An Admin can create a Project, create a Task within it, and assign a Collector — and that Collector sees it and only it.
2	A new Collector can install, onboard, log in, and reach the Dashboard without external help.
3	A Collector can view an assigned Project, its Tasks, and a Task's full instructions/requirements.
4	The Pre-Recording Checklist correctly blocks recording when any condition fails, and correctly allows it when all pass.
5	A full recording session can be captured at 0.5x/0.6x wide-angle, rear-facing, and stopped cleanly.
6	Stopping a recording reliably produces correctly-sized 10-minute chunks, each with a complete, checksum-verified metadata record.
7	Chunks (and their metadata) upload automatically in the background and reach S3 without duplication or corruption.
8	A new recording can start immediately while a previous session is still uploading.
9	Turning off network mid-upload queues the affected chunk(s), and turning network back on resumes upload automatically.
10	A session reaches "Complete" status only once every one of its chunks and their metadata are confirmed uploaded, and the Collector is returned to the Task List/Dashboard.
11	Dashboard and session views accurately reflect pending/uploading/completed chunk counts at all times.
12	An Admin can view and export the metadata for any chunk or session under their Projects.

VOLUME 1 — PRODUCT PLANNING

CHAPTER 1.11

Future Roadmap

What Comes After MVP, and In What Order

Document Title	Future Roadmap
Chapter Reference	1.11
Version	1.1
Status	Draft — Pending Approval
Owner	Faisal — Human Archive
Prepared With	Claude (Cowork)
Date	05 August 2026

Revision Note	v1.1 — Core Admin (create/edit Projects & Tasks, assign Collectors, view status/metadata) moved out of Phase 3 and into MVP. Phase 3 is now scoped to advanced Admin/Ops tooling only (fleet-wide analytics, QA/rework, remote config).
Consistent With	Volume 0, Chapter 0.1, Section 5 (Long-Term Roadmap)

This roadmap sequences work after MVP. Exact dates are intentionally not fixed here — sequencing and dependency order are — timing is set in Volume 11 (Project Management) once pilot data exists.

MVP — Now Includes Core Admin & Metadata

Goal: Ship a two-role platform from day one: an Admin who structures the work, and a Collector who executes it — with every chunk fully and verifiably described by metadata.

- Admin: create/edit Projects and Tasks, assign Collectors, view Session/chunk status and metadata for their Projects.
- Collector: the full confirmed record → chunk → upload flow (Chapter 1.10).
- Full per-chunk metadata capture, integrity verification, and storage — no longer a later enhancement.

Why this changed: The owner determined that a data-collection platform with no way to structure or verify the work isn't a viable first release — Admin and metadata integrity are now foundational, not phase-2 additions.

Phase 2 — Secondary Collector Features

Goal: Reduce Collector friction and add visibility features that don't change the core recording/upload engine.

- Session notes/tagging.
- Task instructions with reference media (images/example clips).
- Push notifications for upload failures and new task assignments.
- Refined multi-project support and UX for switching between concurrent assignments.
- Local caching/offline browsing polish for Projects/Tasks.
- In-app support/help contact.
- Manual pause/resume controls for in-progress uploads (beyond the MVP's manual retry-on-failure).

Dependency: None of these require changes to the core chunking/upload engine — they can begin as soon as MVP is stable in production.

Phase 3 — Advanced Admin & Ops Tooling

Goal: Extend the now-MVP Admin role with fleet-scale visibility, quality control, and remote tunability — the parts of the original Admin Platform concept that go beyond core CRUD and status visibility.

- Live, fleet-wide upload/collection monitoring dashboard across every Project an Admin manages.
- Dataset QA/review and rework-flagging workflow.
- Collector performance and throughput analytics.
- Remote configuration (chunk duration, resolution, bitrate) via feature flags.

Dependency: Builds directly on the Admin role and metadata foundation shipped in the MVP (Chapter 1.6, Chapter 1.3 Section 8) — this phase deepens Admin capability rather than introducing the role itself.

Phase 4 — Future Capabilities

Goal: Extend data quality and platform reach beyond the initial product concept.

- OTA remote configuration of capture parameters without an app-store release.
- AI-assisted in-app QA — blur, occlusion, and motion-blur detection prior to upload.
- Multi-language localization for a global Collector workforce.
- Simultaneous multi-camera capture (front + rear).
- Wearable capture device integration.

Dependency: AI-assisted QA depends on the remote configuration and feature-flag infrastructure built in Phase 3; multi-camera and wearable support depend on the Recording Engine (Volume 5) being re-evaluated for multi-stream capture.

Sequencing Summary

Phase	Primary Value	Blocks / Blocked By
MVP	Admin-structured, metadata-verified, single-Collector record → chunk → upload pipeline	Blocks all later phases
Phase 2	Lower Collector friction, better visibility	Depends on MVP; independent of Phase 3/4
Phase 3	Fleet-scale Admin visibility, QA, remote config	Depends on the Admin/metadata foundation shipped in MVP
Phase 4	Data quality automation, platform reach	Depends on Phase 3 feature-flag infrastructure